

where you keep your site's files). Please note that Apache recommends rewrite rules to be written in the server-wide configuration instead of *htaccess* files if possible, and we're choosing *htaccess* only to keep things simple for the time being.

```
RewriteEngine On
RewriteRule ^([a-z][a-z])/(.+)$ /$2.$1.html [R=307]
```

If you are using `RewriteRule` for the first time, it's a configuration that maps a URL pattern to a transformation. Whenever Apache gets a request from a visitor, it will match the URL against this pattern, and if there is a match, it will rewrite the URL according to the rule. The rule shown above matches every URL that starts with two lowercase letters followed by a forward slash and at least one more character (the dot means any character and the plus means at least one). The second part of the rule (`/($2.$1.html)`) shows the replacement URL, with `$2` and `$1` referring to the second and first parenthesised parts given in the pattern (the tail and the language code, respectively).

Try a redirect first

The third part (`[R=307]`), reserved for flags, currently says the rewrite should happen in the form of a visible 307 (temporary) redirect. We'll change it after testing to make sure the matching and rewriting works. It's easier to test while the rewrite is visible.

Let's see if and how the URLs get rewritten by sending requests using the `curl` command. Alternatively, you can try the developer tools in your favourite browser.



Note: If the above *htaccess* file results in an internal server error, it may mean that you don't have the rewrite module enabled on your server. `sudo a2enmod rewrite` && `sudo systemctl restart apache2` is how you do it on Ubuntu and other Debian-based distros.

Here's what happens with `curl` requests:

```
$ curl -I localhost/en/section/
subsection/page
HTTP/1.1 307 Temporary Redirect
...
Server: Apache/2.4.52 (Ubuntu)
Location: http://localhost/section/
subsection/page.en.html
...
```

Now let's try another language:

```
$ curl -I localhost/ml/section/
subsection/page
HTTP/1.1 307 Temporary Redirect
...
Server: Apache/2.4.52 (Ubuntu)
Location: http://localhost/section/
subsection/page.ml.html
...
```

At this point, a language that you have no translation for will also result in a redirect, but will correctly result in 404 (Not Found) when the page gets redirected. When we make the rewrite an internal process instead of a redirect, incorrect URLs will result in 404 in the first step itself.

Make it internal

Alright, let's make the rewrite internal and invisible to the user. First, make sure you have the files where they are expected to be:

```
~/www$ tree section
section
├── subsection
│   ├── page.en.html
│   └── page.ml.html
```

(In my case, `www/` inside the home directory is the document root; I believe you know where yours is. By the way, you can use a graphical file browser instead of `tree` to check your directory structure.)

Now, how does the *htaccess* file go?

```
RewriteEngine On
RewriteRule ^([a-z][a-z])/(.+)$ /$2.$1.html
```

Yes, the only change was to remove the flags part. Now we can test again to make sure there is no redirect this time:

```
$ curl -I localhost/en/section/
subsection/page
HTTP/1.1 200 OK
...

$ curl -I localhost/ml/section/
subsection/page
HTTP/1.1 200 OK
...
```

Here's another test to make sure languages that we don't have files for result in 404 as expected:

```
$ curl -I localhost/xx/section/
subsection/page
HTTP/1.1 404 Not Found
...
```

Finally, here are tests that actually bring us the page body:

```
$ curl localhost/en/section/subsection/
page
<h1>Welcome!</h1>

$ curl localhost/ml/section/subsection/
page
<meta charset="utf-8">
<h1>സംവാദം!</h1>
```

The same is true from a browser (note how the URLs remain unchanged, although we are still serving *page.en.html* and *page.ml.html*).

Figure 1 shows how English and Malayalam pages get served from *.en.html* and *.ml.html* files while the browser still shows */en/-style* URLs.

Enforcing the new pattern

Alright, the */en/-style* URL works fine, without the user knowing what's under the hood. But the *.en.html* URLs are also accepted without change, if one